

SYSTEMIC REVIEW OPEN ACCESS

A Systematic Review of Published Physiologically Based Pharmacokinetic Models for Drug Excretion Into Human Breastmilk: Knowledge Gaps and Opportunities to Optimize Reporting and Modeling Practices

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ABSTRACT

Physiologically based pharmacokinetic (PBPK) modeling is increasingly used to predict infant exposure to medications via breast milk. Existing guidance on reporting of PBPK modeling analyses is unstructured and nonbinding to researchers. We evaluated reporting practices and modeling approaches in published PBPK modeling studies. A systematic search was conducted across three online databases. Studies published through March 2025 that leveraged PBPK to evaluate milk drug excretion and infant exposure were included. We developed a 22-item structured data abstraction tool—Lactation PBPK Model Reporting Checklist (LAC-PBPK). Study findings were reported using descriptive and narrative analyses. Twenty-six published studies modeling 54 medications met the inclusion criteria. Using the data abstraction tool, only 4 (18%) of reporting domains were reported in all studies. These include the drug modeled, choice of mammary distribution model, context of PBPK modeling (model purpose), and the modeling platform. Most studies (88.4%) implemented whole-body PBPK model structure, with one-half (53%) assuming perfusion-rate-limited milk excretion. Assumptions reflect modeling limitations, primarily due to inadequate reporting in clinical lactation studies and knowledge gaps in infant and mammary physiology. PBPK models can support mechanistic assessment of drug excretion into breast milk, but confidence in model outputs depends on intended use, data availability, and transparent reporting. This highlights the need for a prioritized, fit-for-purpose lactation PBPK reporting framework. From a regulatory perspective, transparent reporting and cautious selection of approaches for estimating infant milk intake, model performance evaluation, physiological extrapolation, and parameter uncertainty analyses are essential to ensure reproducibility and confidence in PBPK-informed infant exposure assessment.

1 | Introduction

Globally, breastfeeding remains the most common and preferred infant feeding modality. The postpartum period, often regarded as the “fourth trimester”, is characterized by

a dynamic and variable restoration of maternal physiology and anatomy to the prepregnancy state [1, 2]. Maternal physiology and infant maturation are significant determinants of infant pharmacokinetics and safety following drug exposure via breast milk [3–5]. Access to robust and reliable guidance is

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Study Highlights

- What is the current knowledge on the topic?
 - Physiologically Based Pharmacokinetic (PBPK) modeling is increasingly being adopted for regulatory and clinical assessment of infant exposure to medications via breast milk. ICH E21 guidelines recommend the use of nonclinical approaches such as PBPK modeling to understand likely drug excretion in breast milk, and subsequent infant absorption and exposure. However, differences in reporting and modeling approaches impede model reproducibility and confidence in PBPK model outputs.
- What question did this study address?
 - The current review evaluated reporting practices and PBPK modeling approaches in the published literature, with a specific focus on understanding breast milk excretion and systemic exposure in breastfed infants.
- What does this study add to our knowledge?
 - Previous reviews have described PBPK modeling of breast milk excretion of environmental chemicals and toxins, as well as veterinary substances. The present review summarizes modeling approaches and reporting practices in lactation PBPK, and proposes a fit-for-purpose prioritization of reporting and study-design elements.
- How might this change clinical pharmacology or translational science?
 - The study highlights reporting differences and the diverse approaches in PBPK modeling, enabling prediction of drug excretion in human milk, thus demonstrating the urgent need for a prioritized, fit-for-purpose approach to lactation PBPK study design and reporting. The study further highlights priority areas (often underreported or not measured) when designing and implementing clinical lactation studies.

important for clinical decision-making when considering the use of medicines during lactation [6]. Concerns about infant safety following maternal medication use are a well-known risk factor for early cessation of breastfeeding [7]. Recent regulatory guidance requires submission of safety evidence for drug therapies of potential use in lactating women [8]. In addition, the ICH E21 guidelines recommend the use of nonclinical studies, such as PBPK models, to provide insight into likely drug levels in breast milk and subsequent infant exposure, absorption, and metabolism [9, 10]. Physiologically based pharmacokinetic (PBPK) modeling is an *in silico* approach that mechanistically characterizes drug disposition. Teorell reported the earliest mechanistic pharmacokinetic modeling in 1937 [11], and PBPK modeling outputs have expanded significantly with advances in computational capabilities [9]. PBPK modeling platforms are increasing; however, limitations in platform-specific modeling workflows, data sources, and reporting affect model reproducibility and confidence in model-informed dosing

recommendations during lactation [12]. Prior to PBPK model development, a feasibility assessment of data sources and reliability is crucial for developing high-quality PBPK models to predict drug excretion in breast milk.

Despite the increasing adoption of PBPK modeling to predict pharmacokinetics during lactation, guidance on the reporting of lactation PBPK model methodology and results is lacking. Two regulatory guidance documents define the scope and content of PBPK analysis reports [13, 14]: the European Medicines Agency (EMA) guidance (first published in 2014 and revised in 2019) and the US FDA guidance (2018). Moreover, the guidance prescribes generic components of a PBPK analysis report, thereby promoting differences in lactation-specific PBPK analysis reports. Two position statements by modeling experts provided templates to foster the use and acceptability of PBPK model reports [15, 16]. The template by Shebley et al. is based on EMA and US FDA guidance, whereas the template by Tan et al. expands on this guidance to address lactational transfer of both therapeutic and environmental chemical substances [15, 16]. A clear understanding of key lactation-specific PBPK reporting components is critical for adequate comparison and interpretation of model-based predictions, thereby supporting the generation of robust evidence to guide pharmacotherapy in lactation.

In summary, existing reporting guidance and templates place little emphasis on reporting of PBPK analyses of breastmilk transfer and infant exposure, and when lactation PBPK modeling is considered, the report focuses on environmental risk assessment, which may not be relevant in pharmacotherapy. A structured review of published models is timely and critical for consolidating understanding of reporting practices and PBPK modeling approaches in human lactation.

2 | Methods

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline (Appendix [17]). The protocol was registered with PROSPERO and can be accessed online using the registration number [CRD42024607862](#). Minor amendments were made to the original protocol's study title and to exclude the risk-of-bias assessment.

2.1 | Search Strategy, Eligibility Criteria, and Data Abstraction

An initial literature search was conducted across three databases (Scopus, PubMed, and Google Scholar) without time restrictions. The keywords used in the search were selected based on a review of several articles on PBPK modeling of drug excretion via breastmilk. The search was conducted by 30th March 2025, covering the titles, keywords, and abstracts of papers using the search terms “PBPK or Physiologically based pharmacokinetic* AND Human AND breast* AND Milk AND (postpartum* OR breastfeed* OR lactat*)”. The review included full-text articles that reported the use of PBPK models to evaluate plasma-to-milk transfer and infant exposure to human pharmaceuticals via breast milk. Duplicate articles and articles published in non-English languages were excluded to ensure

uniformity and ease of data extraction. Eligibility screening of retrieved study articles was conducted by two independent reviewers (J.K. and D.O.). Article screening involved exclusion of articles for which the full text was unavailable. The second screening of article titles and abstracts excluded articles based on a lack of relevance to lactation PBPK modeling, a focus on drug excretion of environmental chemicals via breast milk, preclinical PBPK modeling, and non-primary article types such as commentaries, reviews, editorial opinion pieces, and conference abstracts. Disagreements between the two reviewers (J.K. and D.O.) were adjudicated by a third reviewer (a senior academic supervisor). At the end of the screening process, 26 articles were retained and imported into Zotero Reference Manager (Zotero, Digital Scholar) [18]. Finally, a full-text assessment of the remaining articles was done, and data from eligible studies were abstracted using the LAC-PBPK tool. Data were summarized using descriptive statistics and presented in tables and charts.

2.2 | Lactation PBPK Model Evaluation

A data abstraction tool (LAC-PBPK) was developed to enable consistent evaluation of PBPK modeling approaches and reporting practices. The tool was derived from existing regulatory and consortia reporting guidelines [13–16], reflecting PBPK modeling attributes, such as; study author(s), year of publication, study purpose, sources of model parameter data, model structure, distribution model (e.g., minimal or whole-body PBPK), modeling assumptions, reported study limitations, methods employed in sensitivity analyses, and type of clinical data used for model development. The data abstraction tool was piloted on five published lactation PBPK studies, with iterative amendments of questions to guide data abstraction. The LAC-PBPK tool was developed to meet the study's objectives; however, it can be adapted and improved for future use. The LAC-PBPK tool comprises 22 lactation-relevant PBPK modeling and reporting questions, organized into three domains: “Reporting” (dichotomous responses “Yes/Not reported” to explicitly reported data), “Data” (explicitly reported information), and “Methodological” (modeling approaches) domains. Except for the “Reporting” domain, the remaining two domains were further summarized into descriptive data. The final version of the tool is available as Table S1 (Template data abstraction tool). Descriptive analysis (percentages) and a comprehensive narrative synthesis were used to report numerical data, reported study/drug/PBPK model characteristics, and methodological approaches.

3 | Results

3.1 | Studies Included in the Review

The review evaluated 26 human lactation PBPK studies that modeled a total of 54 medications (Figure 1), with the earliest study published in 2009 [19]. A total of 1279 studies did not meet the eligibility criteria. The majority of studies were excluded for reasons related to PBPK modeling of environmental chemicals excretion in breast milk, preclinical application of PBPK modeling, and secondary article types on milk excretion of drugs.

Of the 26 studies reviewed, 19 (73.1%) developed a PBPK model for a single drug, and the remaining 7 studies evaluated 72.2% of the drugs modeled. The majority (65%) of studies used PBPK modeling in the context of genetic polymorphisms [20–22], integration of nonmechanistic predictive algorithms in PBPK modeling [23–25], infant exposure following maternal use of recommended doses [19, 26–32], mammary drug transporters [33], and fetal exposure contribution in the postpartum period [34]. The remaining studies applied PBPK modeling for bottom-up prediction of drug excretion in breast milk. The majority of studies (96%) used paired (simultaneously collected) maternal plasma and breast milk data to evaluate PBPK model predictions of drug excretion in breast milk. Simcyp population simulator (Certara, UK) [35] was primarily used for prospective simulations (38.5%), and the majority of studies (88.5%) implemented “whole-body” PBPK model structures. More than half (53%) of the studies assumed perfusion rate-limited drug transfer across the mammary epithelium (Table 1).

3.2 | Reporting of PBPK Modeling Results and Approaches

Across the 22-question LAC-PBPK tool, only 4 (18%) questions on the drug modeled, the choice of mammary distribution model, context of PBPK modeling (model purpose), and the modeling platform used were reported in all studies included in the review (Figure 2). Overall, fewer than 50% of studies explicitly reported PBPK model input data for breastmilk parameters, the gestational age modeled, and consideration of mammary transporter kinetics. Evaluation of clinical lactation studies cited in the respective PBPK articles revealed limited documentation of breast milk composition parameters. However, some PBPK modeling studies may not utilize information on infant gestational age at delivery, even when reported in the parent studies [36, 39–40].

3.3 | PBPK Modeling Approaches

Studies included in the review differed in several PBPK modeling approaches (Figure 3). Despite 16 studies reporting a model acceptance criteria metric, only 13 prespecified both the model performance metric and the value/range [20–23, 26–29, 37, 38, 41, 43–44]. Only 7 (26.9%) studies used plasma concentrations to evaluate infant exposure predictions [21, 26, 29, 30, 34, 45]. Except for 4 (25.4%) studies [27, 34, 37, 39], the Relative Infant Dose (RID) metric was primarily used to evaluate pediatric drug exposures via breast milk. Local parameter sensitivity analysis was mostly used to assess model robustness. Most sensitivity analyses were intended to determine the influence of parameter uncertainty [20, 22, 26, 29–31, 37], identification [24, 30, 34, 41, 42, 45], and optimization [38] on pediatric exposure predictions (Table S2).

3.4 | Drugs Modeled

The evaluated PBPK models analyzed the lactational transfer of 54 small molecules from seven therapeutic drug categories, mostly antimicrobials (47%) and central nervous system (CNS) drugs, particularly antidepressants (29%). The majority of drugs modeled

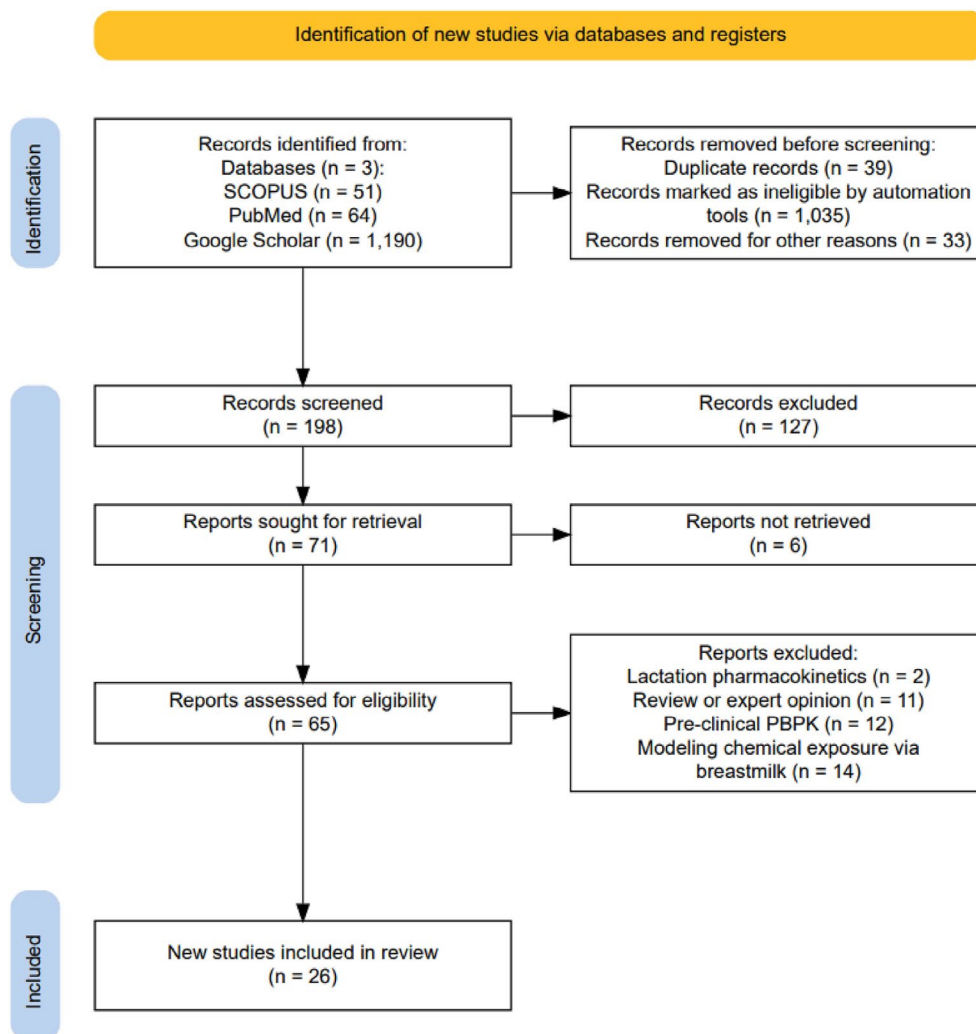


FIGURE 1 | Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [17] flow diagram of physiologically based pharmacokinetic (PBPK) study selection. Of the 1304 articles identified, only 26 met the eligibility criteria. Reasons for exclusion of studies are indicated.

were orally administered (91%), and about half were bases (46.6%) or neutral molecules (32.2%), with potential for ion trapping in a more acidic aqueous milk environment [46]. Maternal renal filtration function decreases (~2-fold) to approximately the prepregnancy state in the first 8 weeks postpartum [47]. In the review, more than half (51.7%) of drugs are cleared via the kidneys, and the remainder via hepatic CYP P450 metabolism (3A4/5—37.9%, CYP2D6—19%), and to a lesser extent, biliary secretion and red blood cell (RBC) esterase enzymes (6.9%). Characterization of drug metabolite pharmacokinetics and its excretion in breast milk may be warranted for some drugs (US FDA) [48]. In the present review, plasma and breast milk concentrations of parent drugs and metabolites are reported for 11 of the modeled drugs [19, 23–25, 28, 29, 32, 36, 37, 41, 44]; however, only three studies modeled both the parent drug and its metabolite [29, 32, 37]. The postpartum changes in mammary drug transporter expression have been described previously [49]. Transporter processes are considered only when passive distribution assumptions and milk properties cannot account for observed breast milk excretion (milk-to-plasma ratios > 1) [50]. In our review, only one [33] out of 8 PBPK studies that modeled potential transporter substrates included mammary transporter contribution.

3.5 | PBPK Modeling Data Sources

PBPK models were parameterized using data from diverse sources (Figure S1). Drug metabolism and clearance were mostly parameterized using in vitro data sources (64% of studies). Breast milk characteristics were largely parameterized using clinical and assumed data sources. Less than one-half of the studies used clinical data to parameterize milk pH [23–25, 41] and creatinine [23–25]. Only 8 (30.7%) studies used clinically measured milk-to-plasma ratios to evaluate milk excretion and infant exposure [20, 23–25, 28, 29, 36, 41]. By comparison, milk parameter values (pH vs. creatinine) applied in modeling ranged 6.5–7.38 vs. 5.5%–7.5% (clinical), 7–7.2 vs. 3%–12% (assumed), and 7.2–7.6 vs. 2%–5% (predicted/software default values). 85% predicted $f_{u,skimmed}$ using software default algorithms.

3.6 | Simulated Versus Observed Postnatal Age

All 26 studies explicitly reported the simulated postpartum period. Reviewing the source lactation pharmacokinetic studies showed differences between the simulated postpartum

TABLE 1 | Summary of lactation PBPK studies evaluated.

Author	Year	Drug(s)	Route	Purpose	Model structure	Maternal data	Infant plasma data	Acceptance criteria Met?	Software
Almond et al. [24]	2025	Antimalarials	Oral	Predictive algorithms	Whole-body	Milk-plasma ^f	No		Simcyp V21
Cole et al. [36]	2024	Albendazole	Oral	Bottom-up prediction	Whole-body	Milk-only	No		Simcyp V23
Ning et al. [20]	2024	Dolutegravir	Oral	Genetic polymorphism	Whole-body	Milk-plasma	No	Yes	Simcyp V22
Yeung et al. [37]	2024	Ezetimibe	Oral	Bottom-up prediction	Whole-body	Milk-only	No	Yes	PK-Sim V11.1
Suryavanshi et al. [34]	2024	Levetiracetam	Oral	Antepartum exposure	Whole-body	Milk-plasma	Yes	Yes	PK-Sim V11.2
Wood et al. [32]	2024	Moxidectin	Oral	Infant safety/risk	Whole-body	Milk-only	No		Simcyp V20.1
Humerickhouse et al. [38]	2024	Pregabalin	Oral	Bottom-up prediction	Whole-body	Milk-plasma-Urine	No	Yes	PK-Sim ^e
Pan et al. [30]	2024	Primaquine	Oral	Infant safety/risk	Whole-body	Milk-plasma	Yes	Yes	Simcyp V21.1
Yeung et al. [39]	2023	Cannabidiol	Oral	Bottom-up prediction	Whole-body	Milk-only	No		PK-Sim V11
Pan et al. [21]	2023	Efavirenz	Oral	Genetic polymorphism	Whole-body	Milk-plasma	Yes	Yes	Simcyp V20.1
Liu et al. [26]	2023	Nilotinib	Oral	Infant safety/risk	Whole-body	Milk-only	No	No	PK-Sim V10
Pressley et al. [40]	2023	Sotalol	Oral	Bottom-up prediction	Whole-body	Milk-only	No		PK-Sim V10
Shenkoya et al. [41]	2023	THC	Inhaled	Bottom-up prediction	Whole-body	Milk-only	No	Yes	Pumas V2.3.2
Damoiseaux et al. [31]	2023	Doxorubicin/ Paclitaxel	IV	Infant safety/risk	Whole-body	None	No		PK-Sim V11.0
Nauwaelerts et al. [28]	2023	Multiple drugs ^a	Oral	Infant safety/risk	Minimal	Milk-plasma ^f	No	No	PK-Sim V9.1
Abduljalil et al. [29]	2022	Theophylline	Oral/IV	Infant safety/risk	Whole-body	Milk-only	Yes	Yes	Simcyp V21
Job et al. [42]	2022	Ondansetron	Oral	Bottom-up prediction	Whole-body	Milk-plasma ^f	No	No	PK-Sim ^e
Pansari et al. [23]	2022	Multiple drugs ^b	Oral	Predictive algorithms	Minimal	Milk-plasma	No	Yes	Simcyp V20
Zhang et al. [33]	2022	Multiple drugs ^c	Oral/IV	Transporter(s)	Whole-body	Milk-plasma	No		Simcyp V21
Yeung et al. [27]	2021	Lamotrigine	Oral	Infant milk intake	Whole-body	Milk-plasma	Yes	Yes	PK-Sim V8
Abduljalil et al. [25]	2021	Multiple drugs ^d	Oral/IV	Predictive algorithms	Minimal	Milk-plasma	No		Simcyp V20

(Continues)

TABLE 1 | (Continued)

Author	Year	Drug(s)	Route	Purpose	Model structure	Maternal data	Infant plasma data	Acceptance criteria Met?	Software
Garessus et al. [22]	2019	Isoniazid	Oral	Genetic polymorphism	Whole-body	Milk-plasma	No	No	R V3.4.1
Olagunju et al. [43]	2018	Efavirenz	Oral	Bottom-up prediction	Whole-body	Milk-plasma	No	Yes	MATLAB V5.1
Delaney et al. [44]	2018	Escitalopram	Oral	Bottom-up prediction	Whole-body	Milk-only	No	Yes	PK-Sim V6.3, MATLAB
Partosch et al. [45]	2018	Ethambutol/ RIF	Oral	Bottom-up prediction	Whole-body	Milk-plasma	Yes		MATLAB V8.0
Willman et al. [19]	2009	Codeine	Oral	Infant safety/risk	Whole-body	Milk-plasma ^f	Yes		PK-Sim V4.0

Note: Antimalarials (**Chloroquine**, **Primaquine**, **Mefloquine**, **Piperaquine**, **Pyrimethamine**, **Atovaquone**, Dihydroartemisinin, Lumefantrine, Proguanil, Pyronaridine, Tafenoquine); NB: Antimalarial drugs with clinical data for model evaluation are represented by **bold font**. Note: Multiple drugs modeled per study.

^aAmoxicillin, Caffeine, Ceftriaxone, Levofloxacin, Metformin, Nevirapine, Sertraline, Valproic Acid, Zidovudine [30].

^bTramadol, Venlafaxine, Fluoxetine, Paroxetine [29].

^cAcyclovir, Bupropion, Cimetidine, Ciprofloxacin, Nitrofurantoin [44].

^dAcetaminophen, Alprazolam, Caffeine, Digoxin [42].

^eThe study did not report the software version used.

^fThe milk-plasma samples were not reported as paired or unpaired.

period and the reported infant age in several cases Figure 4. Simulated postpartum periods and reported infant ages did not overlap in 11.5% of studies [20, 36, 41]. In 61.5% of studies, simulations were conducted beyond the infant age reported in the clinical study (pharmacokinetic extrapolation) [19, 20, 26, 27, 29, 31, 32, 34, 36–39, 41, 44–45], and 27% of studies simulated the postpartum period within the reported infant age [21–25, 28, 33, 40, 42–43]. Overall, only 4 studies (15.4%) simulated within the first 6 weeks, whereas 8 (30.8%), 17 (65.4%), and 24 (92.3%) simulated infant exposure until 24, 48, and 96 weeks postpartum, respectively. Studies by Garessus et al. [22] (isoniazid, 196 weeks) and Willman et al. [19] (codeine, 3 weeks) simulated the longest and shortest postpartum periods, respectively. Across nine (34.6%) studies, postpartum simulations partially evaluated transfer within the first 24 weeks postpartum [19–23, 29, 30, 33, 43], mostly due to a lack of data for model parameterization. Infant growth and maturation of metabolic and clearance processes are prominent in the first half-year after birth [51]. 69.2% [18] of studies simulated infant exposure beyond 24 weeks of life. Except for Garessus et al. [22], most studies incorporated anthropometric and ontogenetic functions when scaling from virtual adults to infant populations over time (Table S2).

3.7 | Lactation-Relevant PBPK Model Assumptions

A total of 23 out of 26 studies reported at least one assumption Figure 5. Only 18 reported assumptions were related to milk characteristics and physiology and were thus considered relevant to lactation PBPK modeling [19, 20, 22–25, 28, 29, 31, 33, 36, 41, 42, 45]. Assumptions include: Equal mammary tissue perfusion rate in term pregnancy and first week postpartum [20, 41], spherical or ellipsoidal mammary acinus/lobule shape [33, 41–42], unidirectional mammary drug transfer [42], fixed infant daily milk intake volume [20, 21, 23–25, 28, 29, 31, 33, 42] and/or frequency [19, 20, 22, 31, 41, 42, 45], sole contribution of passive drug distribution [23, 25, 27, 42], fixed breastmilk pH [25, 36, 41–42] and composition [20, 24, 29], and the absence of drug metabolizing enzymes in human milk [28]. Regarding mammary/breastmilk sub-compartments, researchers assumed nonspecific tissue drug binding [45], homogeneous tissue distribution [41], and negligible drug excretion via the interstitial space [42]. Additionally, equilibration between cord and (maternal) plasma concentrations was assumed when evaluating the contribution of prenatal exposures to postpartum infant pharmacokinetics [34]. To understand the impact of assumed structural/parametric variables on PBPK modeling predictions, sensitivity analyses were performed. Such analyses permit a rational model design and implementation.

3.8 | Lactation-Relevant PBPK Model Limitations

22 out of 26 studies reported at least one limitation to robust PBPK model development (Figure S3). Modeling limitations were related to; (i) Inadequate reporting of clinical lactation study attributes and (ii) Empirical knowledge gaps. Infant breastfeeding status (complementary/exclusive) [34, 41–42], mother/infant CYP phenotypes [23], route of drug administration [23], dosing

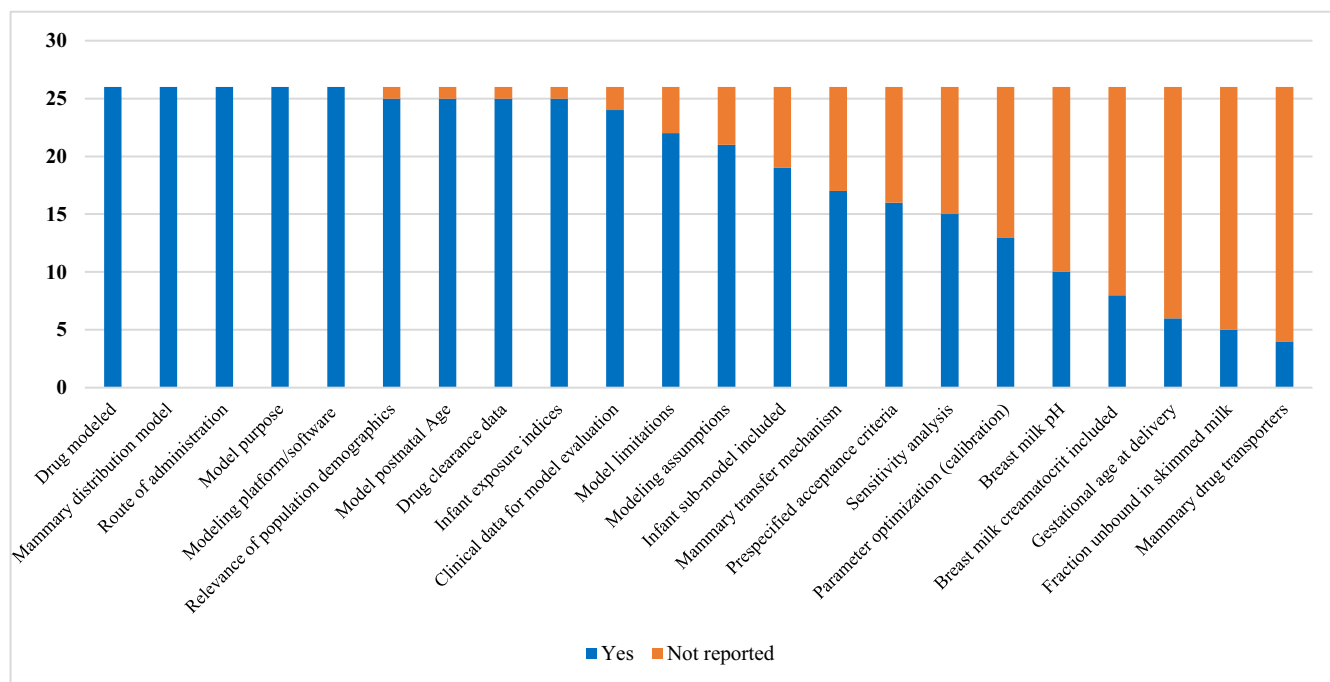


FIGURE 2 | Reporting of PBPK model-related information in published papers. A bar chart to demonstrate descriptive summary “Yes/No” explicit reporting of PBPK model-related information based on a 22-item reporting template (LAC-PBPK).

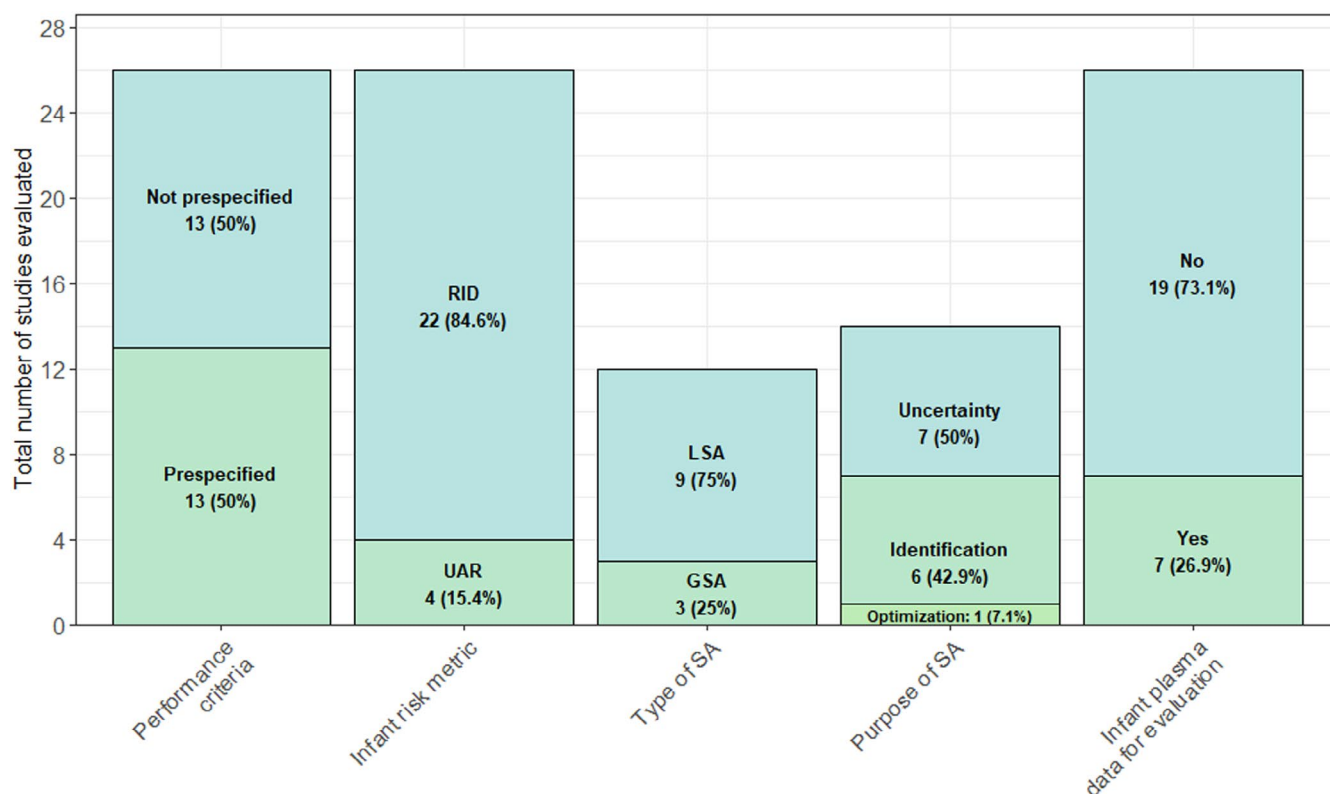


FIGURE 3 | Summary of study modeling feature: (A) whether performance acceptance criteria were prespecified, (B) the infant risk metric used RID (relative infant dose) and UAR (upper area under the curve ratio), (C) the type of sensitivity analysis (SA)—local (LSA) or global (GSA), and (D) the purpose for conducting the sensitivity analysis.

history prior to pharmacokinetic sampling [29], sampling time relative to maternal dosing [27, 28], and maternal food intake relative to dosing [37]. To overcome such limitations, clinical/scientifically plausible modeling assumptions, for example,

exclusive breastfeeding, trough/steady-state concentrations, and clinically relevant doses/formulations, were made. Another common approach was to pool pharmacokinetic observations, regardless of study design or differences in participant

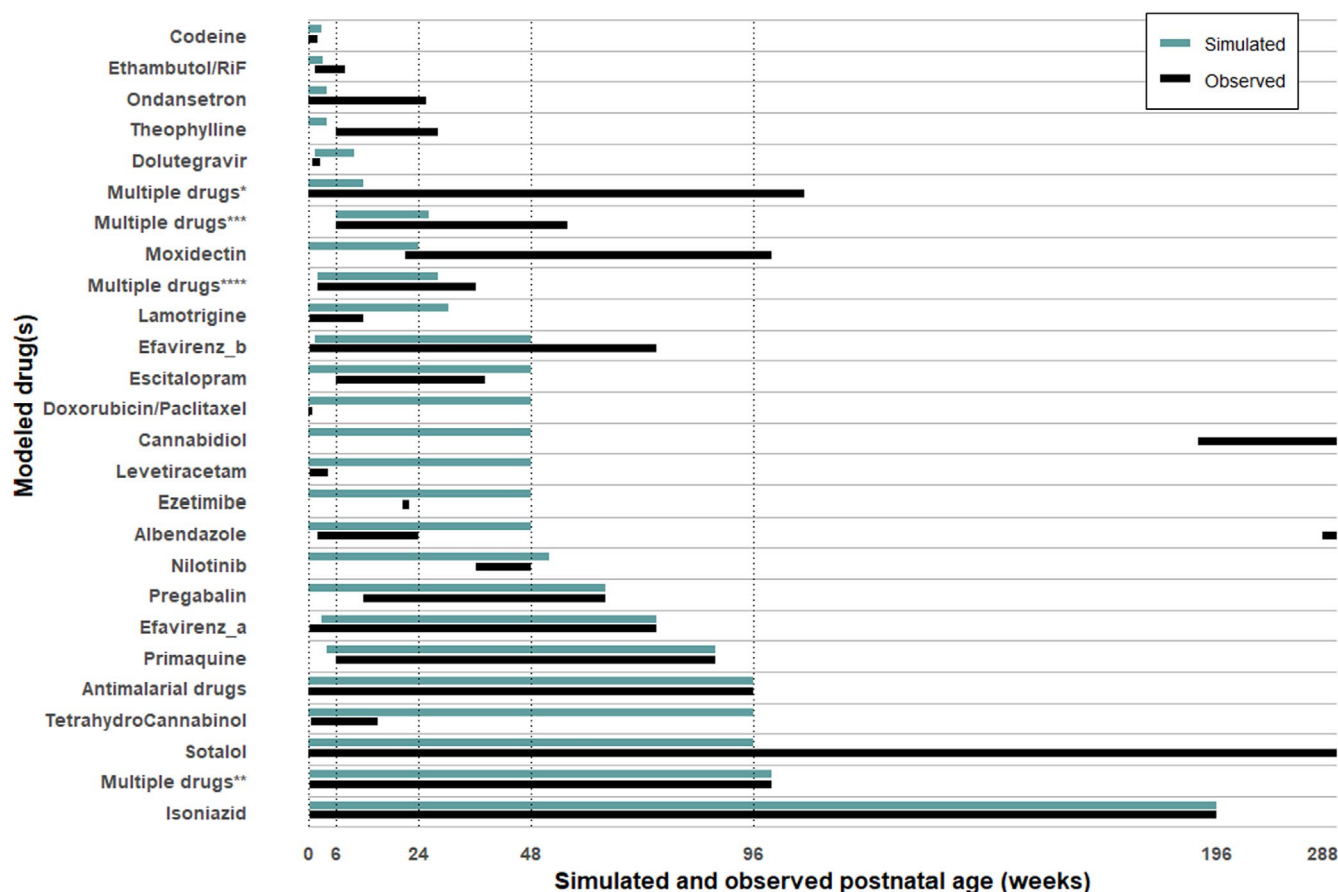


FIGURE 4 | Simulated versus observed postnatal age. Postnatal ages from the source clinical studies (black) versus the ages used in the simulations (teal). Vertical dotted lines (back) indicate relevant postnatal maturational time points. Studies that modeled multiple numbers of drugs were labelled with an Asterix (*): *Amoxicillin, Caffeine, Cetirizine, Levetiracetam, Metformin, Nevirapine, Sertraline, Tenofovir, Valproic Acid, and Zidovudine; **Tramadol, Venlafaxine, Fluoxetine, and Paroxetine; ***Acyclovir, Bupropion, Cimetidine, Ciprofloxacin, and Nitrofurantoin; ****Acetaminophen, Alprazolam, Caffeine, and Digoxin.

characteristics [25, 42] (Table S2). On the other hand, empirical knowledge gaps are mostly related to physiology. These include lactating mammary drug transporter kinetics [22–25, 27, 31, 38, 39, 43], preterm [52] or early-infant-age populations [53], and enzyme ontogeny, especially in polymorphic and non-CYP enzyme systems. To overcome existing modeling limitations, assumptions and sensitivity analyses were employed to increase confidence in PBPK model predictions [19, 33] (Table S2).

3.9 | Lactation-Relevant Model Parameter Sensitivity Analysis and Optimization

Overall, 10 studies conducted sensitivity analyses to identify key parameters driving breast milk excretion, infant exposure, impact of parameter uncertainties on model predictions, and calibrate model parameters using observed data (Figure S3, Figure 2). Model parameters driving model behavior constitute priority parameters for clinical or in vitro measurements to improve confidence in model outputs. Both drug (fraction unbound, lipophilicity, and predicted milk-to-plasma ratio), maternal (breast permeability and milk properties), and infant (binding protein concentration, oral bioavailability, enzyme ontogeny, e.g., Red blood cell esterase) parameters were found to

drive predictions of milk excretion and exposure in breastfed infants (Table S2). To assess the impact of parameter uncertainty on model predictions, sensitivity analyses were performed due to: (1) Inadequate reporting of clinically administered maternal dose, infant milk intake [22], and feeding status in clinical lactation studies (exclusive vs. complementary breastfeeding) [31]; (2) uncertainty around drug properties, for example, specific infant gut permeability, fraction unbound in plasma, and predicted milk-to-plasma ratio [31]; (3) Knowledge gaps related to milk parameters [20], infant oral absorption kinetics [37] and intestinal xanthine methylation enzyme metabolism scalar; (4) explore published milk properties [30] and infant CYP2C8 ontogeny [26] parameter spaces; (5) evaluate worst-case infant systemic and/or local (intestinal) exposures scenarios across a range infant gut maturation characteristics such as transit time, pH, etc. [22, 31]; (6) differences in methods used to estimate tissue partition coefficients (K_p) [31] using the PK-Sim standard method [54, 55] and the Rodgers and Rowland [56] approach. Additionally, 10 studies optimized model parameters, primarily addressing knowledge gaps regarding mammary transfer and infant gut absorption of drugs and their metabolites. Such parameters include: total breast drug clearance [33], breast tissue/milk partition coefficients and scalar values [23, 33, 38], breast tissue flow rate [38], and infant intestinal permeability [21, 27, 28, 39–40].

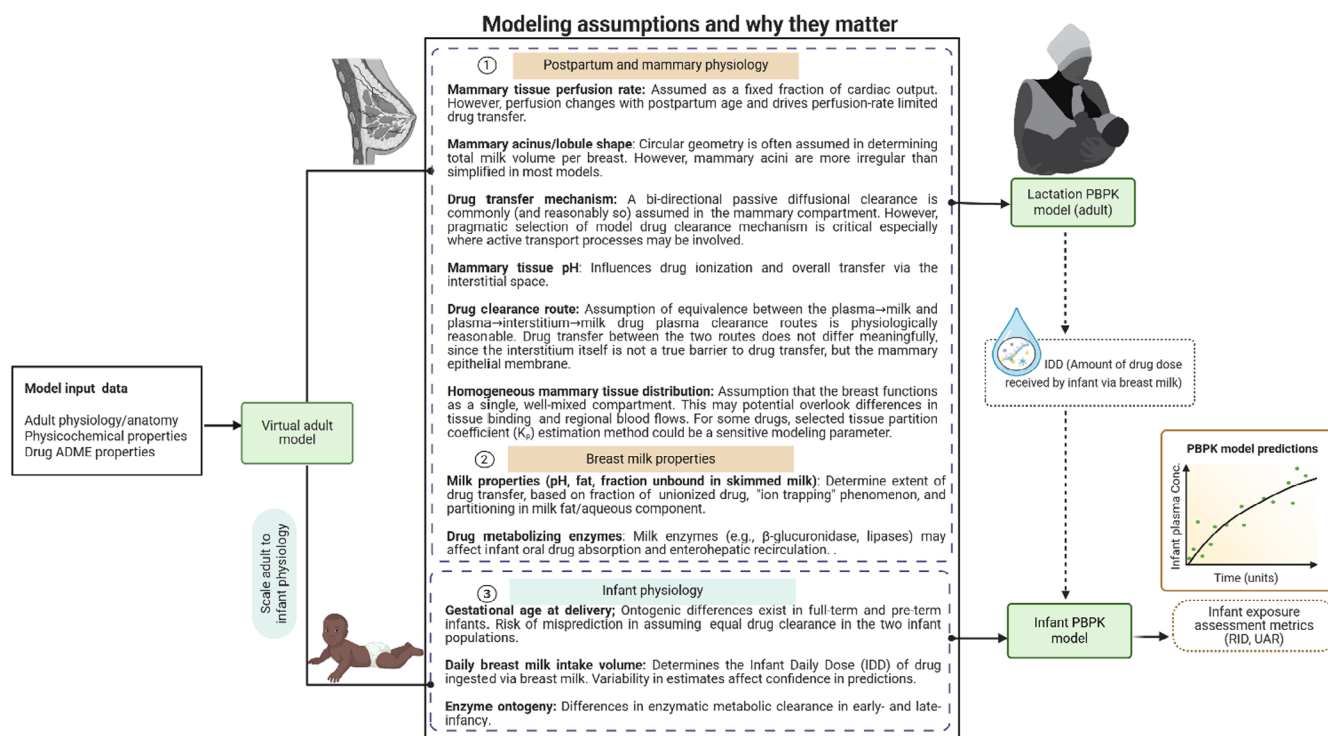


FIGURE 5 | Summary workflow of PBPK modeling: Key modeling inputs, processes, assumptions, and metrics to evaluate breast milk secretion and infant exposure. ADME = absorption, distribution, metabolism, and excretion; pH = concentration of hydrogen ions; RID = relative infant dose; UAR = upper area under the concentration-time curve ratio.

4 | Discussion

The present review appraised 26 publications that applied Physiologically Based Pharmacokinetic (PBPK) modeling to evaluate breast milk drug excretion and infant exposure. We assessed reporting transparency and modeling approaches in PBPK analyses. However, we did not aim to compare or evaluate modeling results across individual studies.

The review demonstrates a rapid adoption of PBPK modeling alongside crucial differences in reporting and mechanistic implementation. Across the 22 LAC-PBPK questions, studies consistently reported PBPK model attributes related to drug distribution, context of use (model purpose), and the software platform used. Transparent reporting of model attributes aligns with recommendations [5], especially since recent findings suggest clinically significant differences in model predictions across software platforms [57]. Conversely, insufficient reporting has downstream effects on PBPK model replication efforts and the subsequent regulatory/clinical relevance of model outputs. While robust PBPK model building is hampered by the scarcity of lactation-relevant clinical data, published articles with limited transparency regarding assumptions and model evaluation approaches further undermine confidence in model outputs. Milk characteristics are often parameterized using data from variable sources (Figure S1). The risk of ignoring drug accumulation in milk remains high under static and mature milk compositional assumptions, which highlights concerns of using weakly basic and lipophilic drugs such as antimalarials and antidepressants [58] during lactation. In some cases, studies neither reported nor specified the infant's gestational age at delivery. Consequently,

the risk of overestimating clearance is high, especially among preterm infants, who exhibit nearly twofold differences in liver metabolism and renal clearance compared with term infants. This presents a safety concern for drugs such as morphine [53, 59] and sotalol [60].

PBPK model fidelity is central to reliable and clinically informative predictions. In the review, modeling assumptions arose in three contexts: limited clinical data (e.g., breast milk parameters); empirical system knowledge gaps (e.g., infant/maternal enzyme ontogenies—CYP 1A2, 2E1, 2D6, and 3A4 in early postpartum); and a parsimonious modeling approach (model simplification, e.g., "lumped" mammary compartments) (Table S2). Explicit reporting of assumptions is essential when dealing with complex physiological systems, enabling easy identification of knowledge and practice gaps. However, rational model simplification is expected to minimize the impact of overlooking system-related differences such as breast milk and mammary tissue sub-compartments. A key component is the mechanism of drug excretion in breast milk. Passive diffusion clearance was assumed for several potential mammary transporter substrates, for example, metformin, albendazole, and paclitaxel [32, 61]. Advancing the field requires adoption of systematic frameworks to foster transparent and consistent consideration of active vs. passive drug distribution assumptions in mammary tissue [62, 63].

Other aspects, including approaches for infant milk intake estimation and model evaluation require urgent consensus within the lactation PBPK community. A reliable infant safety risk assessment combines clinical safety evaluations [64], maternal

TABLE 2 | Fit-for-purpose prioritization of key lactation PBPK study-design and reporting elements.

Element	Priority tier	Why it matters for PBPK utility	Most important when	If unavailable
Maternal dose, formulation, route, dosing history, and sampling time	Minimum essential	These define maternal exposure input and are necessary to interpret maternal plasma and milk concentration data	All lactation PBPK applications	Model input is poorly specified and the interpretation of milk concentrations is weakened
Postpartum timing/infant postnatal age	Minimum essential	Maternal physiology, milk composition, infant intake, and infant maturation change over time	All lactation PBPK applications	Confidence in postpartum extrapolation is reduced
Explicit infant milk intake assumption	Minimum essential	Infant dose from milk depends directly on intake assumptions	All infant exposure predictions	Infant exposure may be over- or underestimated
Transparent model evaluation criteria	Minimum essential	Readers must be able to judge whether model performance is adequate for intended use	All lactation PBPK applications	Fitness-for-purpose cannot be judged clearly
Milk pH and composition	Strongly recommended	These may influence partitioning into milk for compounds where ion trapping or milk partitioning is plausible	Mechanistic interpretation and extrapolative use	Greater reliance on assumptions; reduced mechanistic confidence
Infant gestational age and developmental physiology	Strongly recommended	Infant maturation affects systemic exposure predictions, especially when clearance pathways are immature	Infant systemic exposure predictions, early life, preterm infants	Greater uncertainty in infant exposure and safety interpretation
Sensitivity/uncertainty analysis	Strongly recommended	Helps identify which assumptions drive predictions and where richer data are most valuable	Mechanistic, extrapolative, or higher stakes applications	Model conclusions may appear more certain than justified
Infant plasma data for evaluation	Strongly recommended/conditional	Provides direct evidence on infant systemic exposure rather than inferred dose alone	Models intended to inform infant systemic exposure or risk interpretation	Confidence in infant exposure predictions remains limited
Transporter kinetics	Conditional/advanced	These may be important when passive assumptions do not explain milk transfer or when transporter involvement is plausible	Transporter substrates or unexplained milk transfer behavior	Mechanistic interpretation may be incomplete
Parent–metabolite modeling	Conditional/advanced	Important when metabolites materially influence exposure, activity, or interpretation	Compounds with clinically relevant metabolites	Some relevant exposure pathways may be omitted

Abbreviations: Conditional/advanced, prioritized only for specific compounds, populations, or decision contexts; Minimum essential, expected for most lactation PBPK studies; Strongly recommended, especially important when the model is intended for mechanistic interpretation or extrapolation.

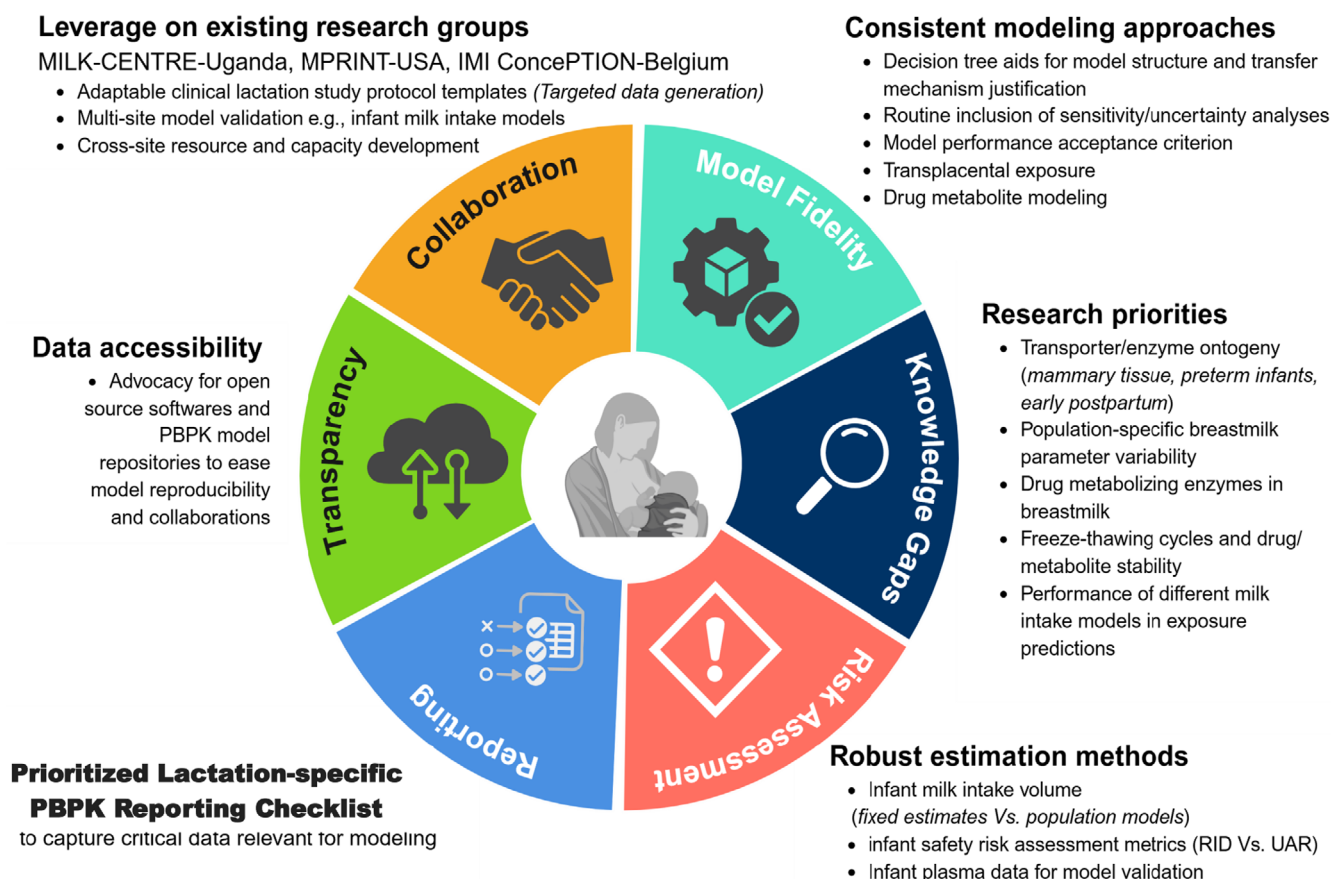
plasma/milk kinetics, and realistic milk intake estimates [27, 65]. Studies relied either on a fixed estimate (US FDA) [55] or a population-based model [27] for milk intake estimation. Both approaches yield comparable estimates only up to 19 days postpartum; thus, the possibility of PBPK overprediction remains a major drawback given changing infant milk intake needs until 6 months postnatal age. Fewer than 25% of studies used infant plasma data, although micro- and sparse-sampling designs exist to ease sample collection for PBPK model evaluation. Furthermore, meaningful assumptions are critical to addressing limitations in PBPK extrapolative capabilities. The absence of ontogeny (preterm vs. early postnatal) or validation data [29, 36, 39] calls for case-by-case extrapolation considerations. Finally, the integration of transplacental exposure into postpartum risk assessment remains underexplored, given its mechanistic complexity and infant/placental delivery dynamics [66, 67].

4.1 | Fit-For-Purpose Application of Lactation PBPK: Why Improved Data and Reporting Still Matter

An important remaining question is whether more complete lactation-specific data and more transparent reporting materially improve the utility of PBPK models, and whether PBPK still adds value when relatively informative empirical lactation data are available. We argue that it does, but in a fit-for-purpose

rather than an absolute sense. Since not all lactation PBPK applications require the same evidentiary standard. Models developed for semi-quantitative screening of infant exposure or for prioritizing compounds for further study may still be informative despite missing some lactation-specific inputs, provided that key assumptions are explicit and the associated uncertainty is clearly described [5, 13–16]. By contrast, applications intended to support mechanistic interpretation, extrapolation across postpartum stages, coupled postnatal-intrapartum infant exposures, evaluation in preterm or developmentally vulnerable infants, or clinical/regulatory decision-making demand greater confidence in biological plausibility, parameterisation, and model evaluation [5, 13–16, 51]. For these uses, more complete reporting and capture of lactation-relevant data strengthen the interpretability and credibility of the model outputs [13–16].

The value of better data extends beyond achieving a closer numerical fit between simulated and observed concentrations. More complete maternal, milk, and infant information reduces reliance on unverifiable assumptions, narrows uncertainty around key exposure drivers, and improves confidence in the mechanism underlying model behavior [12–16]. For example, milk pH and composition may be particularly important when compound properties make ion trapping or differential partitioning into milk plausible [46, 49, 50, 58], whereas infant gestational age at delivery and developmental physiology are especially important when infant clearance is immature or rapidly



Collaborate • Innovate • Standardize • Share

FIGURE 6 | Key considerations in advancing Lactation PBPK modeling.

changing over time [47, 51, 53, 59–60]. In this context, richer data improve both potential predictive performance and the mechanistic defensibility of extrapolation [5, 12, 13, 15].

At the same time, the availability of stronger empirical data does not remove the need for PBPK. Even when maternal plasma and milk concentration data are available, PBPK integrates maternal pharmacokinetics, drug transfer into milk, infant milk intake, developmental changes in infant physiology, and alternative postpartum or dosing scenarios within a single coherent framework [5, 27, 50, 51, 55, 65]. This matters in lactation research because comprehensive empirical data across all relevant postpartum stages, feeding patterns, maternal phenotypes, infant maturational states, and clinically relevant formulations are rare [5, 27, 55, 64–65]. PBPK therefore adds value beyond reproducing observed data by enabling transparent scenario testing, identifying which assumptions drive model behavior, and clarifying where additional measurements would most improve confidence [13–16, 27, 55, 65].

For these reasons, the field would benefit more from a prioritized, fit-for-purpose framework than a rigid universal standard. Some elements should be regarded as minimum essentials for most lactation PBPK studies, whereas others are better treated as strongly recommended or conditionally prioritized depending on the compound, population, and intended use. This approach acknowledges the current maturity of the field while still emphasizing that incomplete documentation can materially limit reproducibility, confidence, and interpretability [13–16]. Based on the patterns identified across the reviewed studies and the considerations discussed above, we propose a fit-for-purpose prioritization of key lactation PBPK study-design and reporting elements (Table 2).

4.2 | Advancing the Field of Lactation PBPK Modeling: Key Considerations

Accordingly, the aim should not be to impose a single maximal standard on all lactation PBPK studies, but to improve transparency and prioritize the data elements most likely to increase confidence for the intended application. To enhance the utility of PBPK modeling in lactation risk assessment, emphasis should be put on the following areas (Figure 6): (1) Adoption of a fit-for-purpose lactation PBPK reporting framework that distinguishes minimum essential, strongly recommended, and context-dependent elements. (2) Promote the use of validated modeling decision trees to support the rational implementation of sensitivity/uncertainty analyses as well as key modeling assumptions. (3) Leverage existing lactation-focused research groups, such as IMI ConcePTION (ended 2024) [68], MILK-CENTRE [61], and MPRINT [69]; to generate evidence on breastmilk intake for infants beyond 6 months and preterm populations, develop adoptable lactation study protocols, and foster cross-population validation of population-based infant breastmilk intake models. (4) Generate knowledge to address gaps in lactation-relevant parameters and aid robust PBPK model development. (5) Advocacy for open-source PBPK software platforms as well as software-agonistic model repositories [32]. This will foster modeling transparency and reproducibility. (6) Diversify disease domain areas for PBPK modeling applications to address emerging

maternal and child health needs, including postpartum-related communicable diseases, for example, gestational diabetes mellitus, hypertensive disorders, etc. (7) Comparison of infant drug exposure estimates based on existing approaches of quantifying infant milk intake.

5 | Limitations of the Review

The review has several limitations. First, it included only studies published in scientific journals up to the last search date; unpublished studies and proprietary models for regulatory submission were therefore unlikely to be captured, which may limit the generalizability of our findings. Second, although comprehensive, the LAC-PBPK reporting checklist captured the key reporting domains, it may require future refinement to accommodate emerging modeling practices. Third, a formal risk-of-bias (quality) assessment was not undertaken, as no standardized appraisal tool currently exists for PBPK studies. Instead, reporting transparency was assessed against established PBPK best-practice recommendations. Consequently, the relative methodological confidence across included studies could not be graded. Finally, the review characterized reporting and modeling approaches rather than evaluating predictive performance, so its findings describe current practice and should not be interpreted as validating the accuracy of any individual model.

Author Contributions

J.K., D.O., F.W.O., A.N.K., A.O., J.K.M., and C.W. wrote the manuscript; all authors designed the research; J.K. and D.O., performed the research; J.K., D.O., K.Y., and N.D., analyzed the data; J.K., D.O., K.Y., N.D., F.W.O., A.N.K., A.O., and C.W. contributed new reagents/analytical tools.

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Conflicts of Interest

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Data sources used in PBPK model parameterization. **Figure S2:** Mammary and breastmilk parameters relevant for lactation PBPK modeling. **Figure S3:** PBPK reporting categories by topic area. The heatmap summarizes how often each reporting category (columns) was documented for each topic area (rows). Numbers are study counts. **Table S1:** LAC-PBPK Reporting Checklist. **Table S2:** Summary of studies included in the review.

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5,6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	5,6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	5,6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g., for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	6
	10b	List and define all other variables for which data were sought (e.g., participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	6

Section and Topic	Item #	Checklist item	Location where item is reported
Effect measures	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results.	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	6
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	N/A
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, meta-regression).	N/A
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	7
Study characteristics	17	Cite each included study and present its characteristics.	7,8
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	N/A
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimates and its precision (e.g., confidence/credible interval), ideally using structured tables or plots.	N/A

Section and Topic	Item #	Checklist item	Location where item is reported
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	N/A
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	15,16,17,18,19,20,21
	23b	Discuss any limitations of the evidence included in the review.	21
	23c	Discuss any limitations of the review processes used.	21
	23d	Discuss implications of the results for practice, policy, and future research.	21,22
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	5
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	2
Competing interests	26	Declare any competing interests of review authors.	2
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	N/A